

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –

RAIGAD -402 103

Winter Semester Examination – December - 2019

Branch:- Mechanical

Sem.:- V

Subject with Subject Code:- Machine Design –I(BTMECE503)

Marks:60

Date:- 13/12/2019

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

(Marks)

Q.1. Any two

- a) Define Machine Design. Explain various design considerations **(6)**
- b) Explain Aesthetic consideration in design with sketches. **(6)**
- c) Explain ergonomics consideration in design of control & display. **(6)**

Q.2. Any Two

- a) The load on a bolt consists of an axial load of 10 kN together with transverse shear force of 5 kN. Find the diameter of bolt required according to maximum shear stress theory. The permissible tensile stress at elastic limit = 100 MPa, and F.S.=1 **(6)**
- b) Cotter joint to support a load of 30 kN. The material used is plain carbon steel with stresses, Tensile stress = 50 MPa, Shear stress = 35 MPa, Crushing stress = 100 MPa Find 1) Diameter of rod, d 2) Diameter of spigot end, d_2 3) Width of cotter, b 4) Thickness of cotter, t 5) Diameter of socket **(6)**
- c) Write design process of knuckle joint with suitable sketches **(6)**

Q.3. Any two

- a) What is stress concentration? What are the causes of stress concentration & remedies for stress concentration. **(6)**

b) Define (any three)

1) Fatigue failure, 2) Endurance limit, 3) Notch sensitivity, 4) Reversed stress
5) Repeated stress (6)

c) A steel rod subjected to reversed axial load of 180 kN. Find the dia. of the rod if F.S. = 2, Ultimate strength = 1070 MPa, Yield strength = 910 MPa, Endurance strength = half of ultimate strength, $K_a = 0.8$, $K_b = 0.85$, $K_c = 0.7$, $K_f = 1$. (6)

Q.4. Any two

a) Find the diameter of solid steel shaft to transmit 20 kW at 200 rpm. The safe shear stress = 45 MPa. If hollow shaft is to be used in place of solid shaft, find the inside & outside diameter when ratio of inside to outside diameter is 0.5 (6)

b) Prove that crushing stress is twice the shear stress when key is equally strong in crushing & shearing and also give sketches. (6)

c) Design muff coupling to transmit 40 kW at 350 rpm. For shaft & key, Shear stress = 40 MPa, Crushing stress = 80 MPa, For muff-Shear stress = 15 MPa. (6)

Q.5. Any two

a) A vertical two start square threaded screw of 100 mm mean diameter and 20 mm pitch supports a vertical load of 18 kN. The axial thrust on the screw is taken by collar bearing of 250 mm outside diameter and 100 mm inside diameter. Find the force required at the end of lever which is 400 mm long in order to lift or to lower the load $\mu = 0.15$ & $\mu_1 = 0.20$. (6)

b) A plate 100 mm wide and 10 mm thick is to be welded to another plate by means of double parallel fillet weld. The plates are subjected to static load of 80 kN. Find the length of weld if shear stress = 55 MPa. (6)

c) Explain with sketch bolt of uniform strength. (6)

Q.6.

a) Define the terms related to springs

- | | |
|-----------------|-----------------|
| 1) Solid length | 2) Free length |
| 3) Spring index | 4) Spring rate. |

(6)

b) Design a helical compression spring for a maximum load of 1000 N for a deflection of 25 mm. Using spring index = 5, Shear stress = 420 MPa, Modulus of rigidity = 84000 MPa, consider effect of stress concentration. (6)

Course: B. Tech. Branch : Mechanical Engineering Semester : V

Subject Code & Name: BTMC 502 Machine Design - I

Max Marks: 60 Date: 31/01/23 Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following. 12

A) Write answers of following multiple choice questions. 1/2 6

- 1) Components made of brittle materials fails because of.....
(A) Sudden fracture (B) Yielding (C) Elastic deflection (D) All of the above
- 2) The following item(s) is (are) produced by the craft evolution process..
(A) Rowing boat (B) Musical instruments
(C) Bullock cart (D) All of the above
- 3) Ergonomics means....
(A) External appearance (B) Natural laws of work
(C) Streamline shape (D) All of the above
- 4) A sunk key fits in the keyway of the
(A) Hub only (B) Sleeve only
(C) Both hub and sleeve (D) Neither hub nor sleeve.
- 5) Woodruff key permits movement between shaft and the hub.
(A) Axial (B) Radial (C) Eccentric (D) None of the listed
- 6) A tension member of 40 mm diameter is to be replaced by a square of same material; side of square will be.....
(A) 40 mm (B) 36 mm (C) 60 mm (D) 25 mm

B) 1) Distinguish between design synthesis and design analysis. 1/2 6

2) What is standardization? Explain its importance in design.

C) Write short note on :- 1/2 6

- 1) Aesthetic considerations in design.
- 2) Factor of safety and its selection

Q.2 Solve Any Two of the following. 12

A) Explain the construction and working principal of cotter joint with the help of neat diagram. 2/2 6

B) The stress induced at a critical point in a machine component made of 45C8 ($S_{yt} = 380 \text{ N/mm}^2$) are as follows, 2/3 6

$$\sigma_x = 100 \text{ N/mm}^2 \quad \sigma_y = 40 \text{ N/mm}^2 \quad \tau = 80 \text{ N/mm}^2$$

Calculate factor of safety by

- 1) Maximum normal stress theory
- 2) Maximum shear stress theory
- 3) The distortion energy theory

- C) A wall bracket with a rectangular cross section is as shown in the Figure 1, the depth of the cross section is twice that of the width. The force P is acting on the bracket at 60 degree to the vertical is 6 kN. The material is made of Gray cast iron FG 250 and the factor of safety is 3. Determine the dimensions of the cross section of the bracket. Assume maximum normal stress theory of failure.

2/4

6

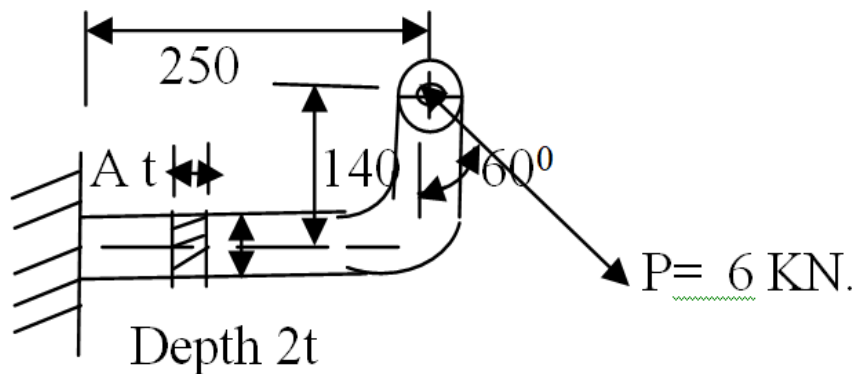


Figure 1. (All dimensions are in mm)

Q. 3 Solve Any Two of the following.

12

- A) 1) What is the stress concentration? Explain the various methods of reducing stress concentration.
2) What is fluctuating stress? Draw a stress – time curve for fluctuating stress.
3) Explain the terms fatigue failure & fatigue life?
- B) A plate made of steel 20C8 ($S_{ut} = 400 \text{ N/mm}^2$) in hot rolled and normalized condition is as shown in the Figure 2. It is subjected to completely reversed axial load of 30 kN. The notch sensitivity factor can be taken as 0.8 and the expected reliability is 0.85. The factor of safety is 2. Determine the plate thickness for infinite life. Use the below data if required
 $K_a = 0.67$, $K_b = 0.85$, $K_c = 0.897$, $K_t = 2.51$

1/2

6

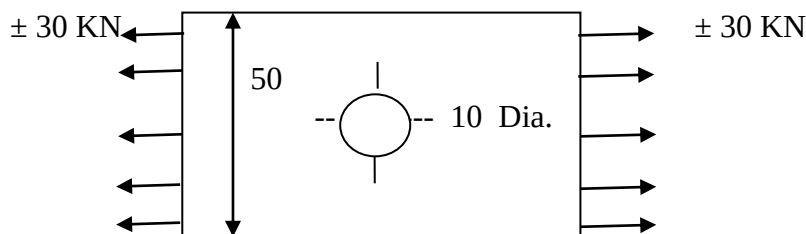


Figure 2. (All dimensions are in mm)

- C) A transmission shaft of cold drawn steel 27Mn2 ($S_{ut} = 500 \text{ N/mm}^2$ and $S_{yt} = 300 \text{ N/mm}^2$) is subjected to a fluctuating torque which varies from -100 N-m to + 400 N-m. If Factor of safety is 2 and expected reliability is 90 %. Neglecting the effect of stress concentration, determine the diameter of shaft assuming distortion theory of failure. Use the below data if required $K_a = 0.79$, $K_b = 0.85$, $K_c = 0.897$

2/5

6

Q.4 Solve Any Two of the following.

		12
A)	What is the function of keys? Explain the different types of keys along with their neat sketches.	1/2 6
B)	A shaft made of plain carbon steel 40C8 ($S_{yt} = 380 \text{ N/mm}^2$) is mounted with pulley 1000 mm in diameter overhang the left hand bearing by 250 mm. The belts mounted on it transmit the power to machine shaft below the pulley. Tension on tight side and slack side are 3 KN and 1 KN respectively. The weight of pulley is 500 N. The angle of wrap is 180 degree. If factor of safety is 3, determine the diameter of the shaft.	3/6 6
C)	A flexible coupling is used to transmit 15 KW at 100 rpm. There are 6 pins at pitch circle diameter (PCD) 200 mm. the effective length of the bush is 35 mm. the gap is 5 mm and pin length is 23 mm. The permissible shear and bending stress for the pin are 35 N/mm^2 & 152 N/mm^2 respectively. Calculate the pin diameter under shear and bending consideration.	2/6 6

Q. 5 Solve Any Two of the following.

		12
A)	1) What is the cause of residual stresses in welded joint? How are they relieved? 2) Explain the surge phenomena in spring and methods to avoid the same.	1/2 6
B)	A helical compression spring made of circular wire is subjected to an axial force which varies from 2.5 KN to 3.5 KN. Over this range of force, the deflection of the spring is 5 mm .the spring index can be taken as 5. The spring has square and ground ends. The spring is made of cold drawn steel wire with ultimate tensile strength of 1050 N/mm^2 and modulus of rigidity of 81370 N/mm^2 . The permissible shear strength for the spring wire should be taken as 50 % of the ultimate tensile strength. Design the spring and calculate a) Wire diameter, b) Mean coil diameter, c) Number of active coils, d) Total number of coils. e) Solid length of the spring, f) Free length of the spring.	3/6 6
C)	A single start square threaded screw is used in a screw press to exert a force of 75 KN. The screw is made of plain carbon steel 10C4 having ultimate strength = 350 N/mm^2 and factor of safety of 4. The permissible compressive stress is equal to permissible tensile stress and permissible shear stress is 50% of permissible tensile stress. The nut is made of grey cast iron FG200 and permissible bearing pressure between contacting surfaces of screw and nut is 17 MPa. A low friction thrust ball bearing is used in the mechanism and collar friction can be neglected. The coefficient of friction at the thread surface between steel screw and cast iron nut can be taken as 0.15. Determine the size	3/6 6

of screw and length of nut. The axial length of threads in the nut should be between 1- 1.5 times the nominal diameter of the screw. (Use the below data)

Sr. No	Nominal diameter, d (mm)	Pitch, p (mm)
1	30,32, 36	6
2	40 , 44	7

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: T.Y. B. Tech. Branch: Mechanical Semester: V Subject Code & Name: BTMEC503 (Machine Design I) Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Define and explain Fatigue failure, Endurance limit, Notch sensitivity	CO2	6
B)	Explain the Rankin's theory of failure	CO2	6
C)	Explain Maximum shear stress theory	CO3	6
Q.2	Solve Any Two of the following.		
A)	Assume the data of the transmission shaft given in Example 9.5. For this shaft, the permissible angle of twist is 3°/meter length. The modulus of rigidity of the shaft material is 79300 N/mm ² . Calculate: (i) the permissible angle of twist; and (ii) the shaft diameter on the basis of torsional rigidity.	CO2	6
B)	A solid shaft is transmitting 1 MW at 240 rpm. Determine the diameter of the shaft if the maximum torque transmitted exceeds the mean torque by 20%. Take the maximum allowable shear stress as 60 MPa	CO1	6
C)	Explain Aesthetic and ergonomic consideration in design	CO3	6
Q. 3	Solve Any One of the following.		
A)	Explain with diagram design procedure of knuckle joint	CO3	12
B)	What are the different types of springs? Explain with neat sketch.	CO2	12
C)	Explain Design procedure for Muff Coupling.	CO2	12
Q.4	Solve Any Two of the following.		
A)	Illustrate design procedure for square key	CO1	6
B)	What is a butt weld joint? Discuss design of butt weld joint under tension.	CO2	6
C)	Write note on Creep.	CO1	6

Q. 5	Solve Any One of the following.		
A)	It is required to design a helical compression spring subjected to a max. Force of 7.5 kN. The mean coil diameter should be 150 mm from space consideration. The spring rate is 75N/mm. The spring is made up of oil hardened and tempered steel with ultimate tensile strength of 1250 N/mm ² . The permissible shear stress for the spring wire is 30% of the ultimate tensile strength ($G = 81370 \text{ N/mm}^2$) Calculate: (1) Wire Diameter (2) No. of active coil.	CO3	12
B)	A rigid Coupling is used to transmit 50 kw power at 300 rpm. There are 6 bolts. The outer diameter of the flanges is 200 mm, while the recess diameter is 150 mm. The bolts are made of steel 45C8 ($S_{yt} = 380 \text{ N/mm}^2$) and the factor of safety is 3. Determine the diameter of the bolts. Assume the bolts are fitted in large clearance hole.	CO2	12
	*** End ***		

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Supplementary Summer Examination – 2023

Course: B. Tech.

Branch : Mechanical

Semester :V

Subject Code & Name: BTMC502 – Machine Design - I

Max Marks: 60

Date:09/08/2023

Duration: 3 Hr.

Instructions to the Students:

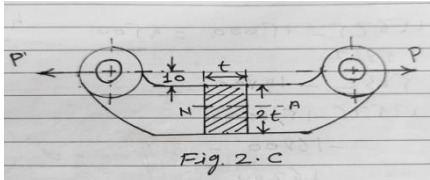
1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.	12
A) Describe Machine Design Procedure.	6
B) Explain Basic type of product form in Aesthetics design	6
C) Explain with sketch man-machine closed loop system.	6
Q.2 Solve Any Two of the following.	12
A) A cylindrical shaft made of steel of yield strength 700mpa is subject to static load consisting of B.M.10 KN-m and a tortional moment of 30 KN-m. Determine the diameter of shaft using maximum shear stress theory of failure. (F.S. = 2, E = 210 Gpa and Poission's ratio = 0.25)	6
B) Design Cotter joint. P = 30 KN Material = Carbon Steel Shear Stress = 35 N/mm ² Tensile Stress = 50 N/mm ² Compressive Stress = 90 N/mm ²	6
C) Explain design procedure of Knuckle joint.	6
Q. 3 Solve Any Two of the following.	12
A) Define any two- Stress Concentration and its remedies. Fatigue failure and Endurance limit Reversed Stress and Repeated Stress	6
B) Explain – i) Goodman dia ii) Solderberg dia	6
C) A steel rod subjected to a revered axial load of 180KN. Find the diameter of the rod. , If F.S= 2 $\sigma_u = 1070\text{MPa}$ $\sigma_y = 910\text{ MPa}$ $\sigma_e = \frac{1}{2} \sigma_u$	6
Factors- $K_a = 0.8$ $K_b = 0.85$ $K_c = 0.7$, $K_f = 1$	

- Q.4 Solve Any Two of the following.** **12**
- A)** Prove that for square key crushing stress is twice the shear stress. **6**
- B)** Design muff coupling which is used to connect two shaft transmitting 40KW at 350rpm **6**
- Material for Shaft and Key = Plain Carbon Steel
 τ = Shear Stress = 40 Mpa
 σ_c = Crushing Stress = 80 Mpa
 Material for Muff is C.I
 τ = Shear Stress = 15 Mpa
- C)** Explain design procedure of Bushed-Pin type flexible coupling. **6**
-
- Q. 5 Solve Any Two of the following.** **12**
- A)** Explain Bolt of uniform strength **6**
- B)** A plate of 100mm wide and 10mm thick is to be welded to another plate by means of double parallel fillet weld. The plates are subjected to a load of 80 KN. Find the length of weld if shear stress = 55 Mpa. **6**
- C)** Design a helical compression spring for a maximum load of 1000 N for a deflection of 25mm. **6**
- C = Spring Index = 5
 Shear Stress = 420 N/mm²
 G = 84 KN/ mm²
 Consider Wahl's Factor

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination-2023 Course: B. Tech. Branch : Mechanical Engineering Semester : V Subject Code & Name: BTMEC 503 Machine Design –I Max Marks: 60 Date: 11/8/23 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Write answers of following MCQ'S 1) The following standard(s) is (are) used in Mechanical engineering design. a) Standards for materials, (b) Standard for testing of products (c) Standards for Fits, Tolerances and surface finish of component (d) All of the above. 2) Aesthetic considerations in designs are related to--- a) Colour combinations, b) Curved surfaces, c) Easy to handle d) External appearance of the product 3) A Splined shaft is one having---- a) Single Key B) Multiple Keys C) Keys as a integral part of shaft 4) For cast iron components -----is considered to be the failure criterion. a) Ultimate tensile strength , b) Yield Strength, c) Endurance limit d) None of the mentioned 5) A beam subjected to bending moment undergoes the following stresses. a) Compressive , b) Tensile, c) Both compressive & tensile d) None of the mentioned 6) Determine the width of the cotter used in cotter joint connecting two rods subjected to axial load of 50kN and permissible shear stress in cotter is 50N/(mm ²). Given thickness of cotter=10mm a) 50 mm , b) 100 mm, c) 150 mm d) 25mm	1/1	6
B)	Explain the term stress concentration and the methods to reduce the same.	1/2	6
C)	Explain the aesthetic and ergonomic consideration in design & its importance .	1/2	6
Q.2	Solve Any Two of the following.		12
A)	Explain the construction and working principle of Knuckle joint with the help of neat diagram.	2/2	6
B)	A cotter joint is to be used to connect two rods made of plain carbon steel 30 C8 having $S_{yt} = 400 \text{ N/mm}^2$ & subjected to a tensile load of 125 KN. The cotter is made from steel plate with thickness of 15 mm. Calculate the dimensions of socket and cotter assuming following data:- The yield strength in compression is twice the yield strength in tension. The yield strength in shear is 50% of the yield strength in tension. The factor of safety is 5.	2/3	6
C)	An offset link subjected to a force P of 30 KN is as shown in the fig. 2 c. it made of cast iron FG300 having $S_{ut} = 300 \text{ N/mm}^2$ and the F.S is 3 . Determine the dimensions of cross section of the link.	2/4	6
			

Q. 3	Solve Any Two of the following.		12
A)	Explain the following terms:- 1) Notch Sensitivity 2) Endurance limit and factors affecting it.	2/2	(2X3)=6
B)	A rotating bar made of steel 40 C8 having $S_{ut}=600 \text{ N/mm}^2$ is subjected to completely reversed bending stress. The corrected endurance limit of the bar is 320 N/mm^2 . Calculate the fatigue strength of the bar for a life of 90,000 cycles.	2/5	6
C)	A spherical pressure vessel having inner diameter of 500 mm is welded from steel plate. The welded joints are sufficiently strong. The plate is made from cold drawn steel 20C8 ($S_{ut} = 400 \text{ N/mm}^2$ and $S_{yt} = 245 \text{ N/mm}^2$). The vessel is subjected to internal pressure varying from zero to 6 N/mm^2 . The expected reliability is 50 % and F.S. is 3. The size factor is 0.85. the vessel is expected to withstand infinite no of stress cycles. Calculate the thickness of the plate. Use the following data:- $K_a = 0.82$, $K_b = 0.85$, & $K_c = 1.0$	3/3	6
Q.4	Solve Any Two of the following.		12
A)	what is the function of key? Explain its types with the help of neat diagram.	2/2	6
B)	A shaft 50 mm in diameter is used to transmit 30 KW power at 250 rpm to the gear mounted on it. A square key is used to connect the gear on shaft. The key is made of steel 50 C4 having $S_{yt} = 450 \text{ N/mm}^2$ and the F.S is 3. For the key material yield strength in compression can be assumed to be equal to the yield strength in tension. Determine the dimensions of the key.	2/6	6
C)	A rigid flange coupling is used to transmit 25 KW power at 750 rpm. There are 4 bolts at pitch circle diameter of 125 mm. The shaft and bolts are made of 40 C 8 ($S_{yt} = 400 \text{ N/mm}^2$) and factor of safety is 3. Assume $S_{sy} = 0.577 S_{yt}$ and determine the diameter of shaft and bolts.	3/6	6
Q. 5	Solve Any Two of the following.		12
A)	Explain the following terminology used in springs with the help of neat diagrams :- 1) Free length 2) compressed length 3) solid length	2/3	(3X2) 6
B)	A tension spring balance consists of helical tension spring which is attached to fixed support at one end and carries masses at free end. The pointer attached to free end moves on a scale and indicates mass. The length of the scale is 100 mm which is divided into 50 equal divisions. Each division on scale indicates 0.5 kg . The max. Capacity of the balance is 50 kg. and the spring index is 6 . The spring is made of oil hardened and tempered steel wire of grade SW ($G = 81370 \text{ N/mm}^2$ & $S_{ut} = 1600 \text{ N/mm}^2$) . The permissible shear stress in the spring wire is 50 % of the ultimate tensile strength. Design the spring and give its specification.	2/5	6
C)	The lead screw of a lathe machine is provided with Acme threads having 50 mm outside diameter and 8 mm pitch. The axial force exerted by lead screw is 2500 N. The collar subjected to thrust in the carriage has 110 mm outside diameter and 55 mm inside diameter and the lead screw rotates at 30 rpm. The coefficient of friction for screw is 0.15 and that for collar is 0.12 . Determine 1) The power required to drive the screw and 2) The efficiency of the lead screw.	2/4	6
	*** End ***		

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech Branch : Mechanical/Mechanical(Sandwich) Semester : V

Subject Code & Name: BTMC502 Machine Design – I

Max Marks: 60

Date:03-01-24

Duration: 3 Hr.

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Use of Design data book is allowed.*
5. *Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Write short note on preferred sizes of component and make the table for all the series.		6
B) Explain basic Procedure of Machine Design		6
C) Which are the basic type of product forms.		6
Q.2 Solve Any Two of the following.		12
A) A mild steel shaft of 50 mm diameter is subjected to a bending moment of 2000 N-m and a torque T. If the yield point of the steel in tension is 200 MPa, find the maximum value of this torque without causing yielding of the shaft according to 1. The maximum principal stress		6
B) How to design the Knuckle joint		6
C) Design a cotter joint to support a load varying from 30 kN in compression to 30 kN in tension. The material used is carbon steel for which the following allowable stresses may be used. The load is applied statically. Tensile stress = compressive stress = 50 MPa ; shear stress = 35 MPa and crushing stress = 90 MPa.		6
Q. 3 Solve Any Two of the following.		12
A) State and explain stress concentration factor and its causes and remedies		6
B) Explain Goodman diagram.		6
C) Explain Reversed Stresses-Design for Finite and Infinite Life		6
Q.4 Solve Any Two of the following.		12
A) A solid circular shaft is subjected to a bending moment of 3000 N-m and a torque of 10 000 N-m. The shaft is made of 45 C 8 steel having ultimate tensile stress of 700 MPa and a ultimate shear stress of 500 MPa. Assuming a factor of safety as 6, determine the diameter of the shaft.		6
B) Explain spline keys		6
C) Design a cast iron protective type flange coupling to transmit 15 kW at 900 r.p.m. from an electric motor to a compressor. The service factor may be assumed as 1.35. The following permissible stresses may be used:		6

Shear stress for shaft, bolt and key material = 40 MPa

Crushing stress for bolt and key = 80 MPa

Shear stress for cast iron = 8 MPa

- Q. 5 Solve Any Two of the following.** **12**
- A) Explain Different forms of threads. **6**
- B) State the Advantages and disadvantages of welded joints. **6**
- C) A helical spring is made from a wire of 6 mm diameter and has outside diameter of 75 mm. If the permissible shear stress is 350 MPa and modulus of rigidity 84 kN/mm^2 , find the axial load which the spring can carry and the deflection per active turn. **6**

***** End *****